

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Number of units = 7×8 (1) Cost = $56 \times 30 = 1680$ (1) = [£]16.80 (ecf) (1) ecf must apply to a power $\times$ time Alternative: 30×7 (1) = $210 \times 8 = 1680$ (1) = [£]16.80 ecf (1) ecf must apply to a cost for 1 hour calculation Alternative: 30×8 (1) = $240 \times 7 = 1680$ (1) = [£]16.80 ecf (1) ecf must apply to a power of 1 kW calculation	1	1 1		3	3	
	(b)	(i)		Requires 16 litres i.e. $\frac{240}{15}$ (1) Cost = $1.60 \times 16 =$ [£]25.60 (1)		2		2	2	
		(ii)	I	$\frac{14400}{240} = 60$ (1) 25.60 ecf – 16.80 ecf = 8.80 so Difference in cost/y = £8.80 ecf $\times$ 60 ecf = [£]528 (1) Alternative: Petrol costs 25.60 ecf $\times$ 60 ecf = 1536 and electricity costs 16.80×60 ecf = 1008 (1) Difference = [£]528 (1)		2		2	2	

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			II	Extra cost of car = £12 000 (1) Payback time = $\frac{12\,000}{528\text{ ecf}}$ = 22.7 or 22.73 or 23 [years] (1) Don't accept 22.8 [years]		2		2	2	
	(c)			{Electricity / energy} must be used [for charging] (1) For the 2nd mark EITHER Likely to come from a fuel power station [which emits greenhouse gases] - so disagree OR comes from renewable or nuclear energy [which do not emit greenhouse gases] - so agree Conclusion must be present to award 2 marks			2	2		
				Question 4 total	1	8	2	11	9	0